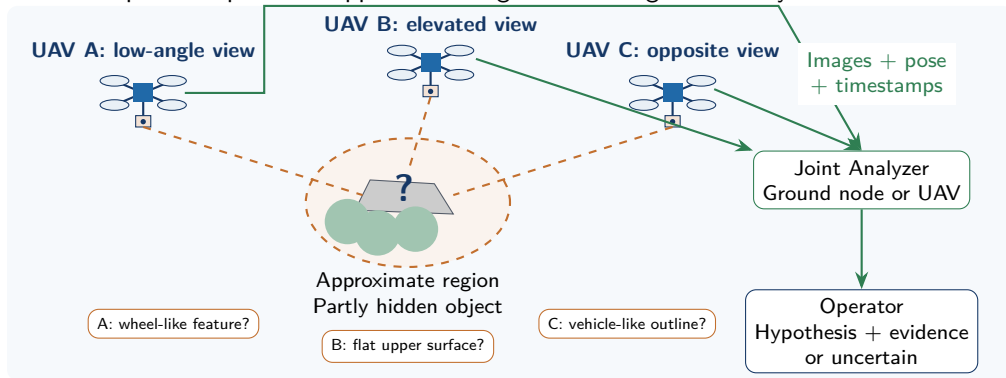


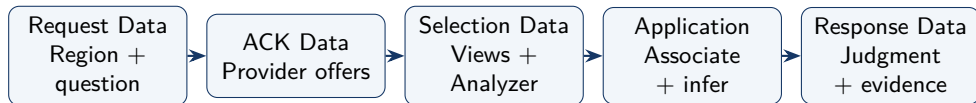
What Is Near This Map Pin? Combine Partial Views

Request: inspect this approximate region. The target's identity is unknown.



Evidence: Illustrative clues, not measured detections. Orange: sight lines; green: named-data retrieval. No individual view is assumed sufficient.

Complementary Evidence Supports a Joint Judgment



- ▶ **Cameras:** receive a region and viewpoint, without a target class.
- ▶ **Analyzer:** match the same object across views and combine complementary cues.
- ▶ **NDNSF:** select Providers, assign roles and retrieve protected evidence.

Key point

Multiple views can reduce ambiguity. Multiple UAVs add parallel capture; compare against one UAV changing viewpoint.

Evidence: Proposed workflow. Conflicting or insufficient evidence yields uncertainty. Three cameras are illustrative.

Three Gaps Between Simulation and the Field

Gap	Simplified example	Real deployment needs
Position / pointing	Known coordinates; assigned views	GPS + IMU, camera-to-body calibration, coordinate frames, synchronized pose and exposure
Useful views	Target assumed visible	Region search, camera control, visibility checks and complementary viewpoints
Joint judgment	Views assumed to match	Same-object association, time alignment, multi-view inference and uncertainty checks

Key point

Nearby images may show different objects. Validate cross-view association and joint judgment against ground truth.

Evidence: Candidate tools: MAVSDK control/telemetry; OpenCV calibration/geometry. Hardware support and integration remain to be verified.

Optional Gimbal Upgrade: Not the Current Setup

SIYI A8 mini
SIYI



- ▶ Camera + three-axis gimbal in one unit.
- ▶ Ethernet UDP / UART control; Ethernet video.

Gremsy Pixy U



- ▶ Gimbal only; add a compatible camera.
- ▶ SDK / MAVLink control; verify camera support.

Current choice: fixed camera. Gimbal hardware is a possible later upgrade.

Evidence: Product images: SIYI official store and Gremsy official site. Images not to scale. Candidate hardware; verify power, payload and mounting before selection.

Current Solution: Fixed Camera

Why start here?

Simpler hardware; constant camera-to-body geometry.

- ▶ **Calibrate:** camera intrinsics and the fixed camera-to-body transform.
- ▶ **Capture:** reposition the UAV; prefer low-motion intervals and short exposures when feasible.
- ▶ **Record:** pair each image with capture-time position, attitude and timestamp.
- ▶ **Analyze:** estimate viewing direction from attitude and calibration; verify same-object matches before combining views.

Limitation

Attitude estimates do not prevent blur or jitter. Reject poor frames.

Evidence: Implementation plan; calibration and timing require validation. Metadata reference: MAVLink capture metadata.